# T14 Top-Down Slice

## CRE132 worksheet

### What you'll make

A small dungeon of four rooms that uses everything from the top-down samples: movement, sword and fireballs, slimes, pickups and the inventory, a locked door, a pressure plate and a lever, an old adventurer who gives hints, cameras that blend from room to room, and chests.

There is **no new code** in this sample. You'll take a guided tour of how the level is put together, finding each earlier sample's piece in the Hierarchy, then extend the level yourself: a tougher slime, a second key and door, a new room.

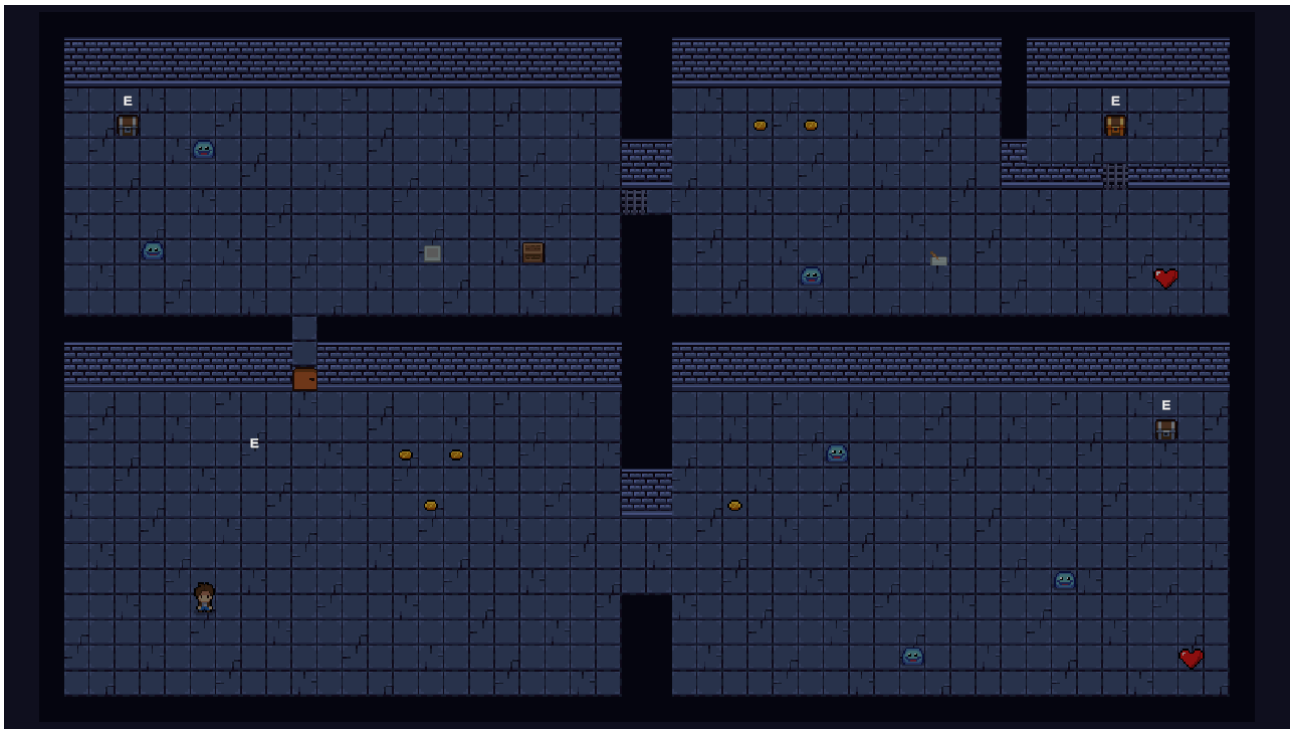


Figure 1: The whole dungeon from above: four rooms with slimes, chests, keys, doors and hearts.

### Before you start

There is no Starter project for this sample: you work in the Finished one.

1. Unzip `CRE132_T14_TopDownSlice_Finished.zip`. In Unity Hub, choose **Add > Add project from disk** and pick the `CRE132_T14_TopDownSlice_Finished` folder. Open it with Unity **6000.3.11f1**.
2. Open the scene `Assets/_game/Scenes/TopDown/T14_TopDownSlice`.
3. Press **Play**. Walk with WASD or the arrows, swing the sword with **J** or the left mouse button, hold the right mouse button to throw fireballs at the pointer, and press **E** to talk, pull levers and open chests.

### 1. Play it through

Play the whole dungeon once before you look inside. The route:

- **Room A (entrance):** talk to the **Old Adventurer** and pick up the coins. The way up is a **Locked Door**.

- **Room B (east):** three slimes, a heart, and a chest with the **Gold Key**.
- Back through A and through the door to **Room C (north-west):** two slimes, a chest with a potion, and a gate to the east held open by a **Pressure Plate**. Standing on the plate opens the gate, but it shuts when you step off. Push the **Crate** onto the plate instead.
- **Room D (north-east):** a slime, a heart, coins, and a **Lever** that opens the alcove with the **Treasure Chest** (5 blue gems).

Every Play starts fresh: nothing is saved between plays here (part 2 explains why).

## 2. The tour: how this level is put together

Stop Play and look at the Hierarchy. Each group of objects comes from an earlier sample. Select the objects named in bold and look at their components in the Inspector.

**Grid** (C03, T03). The **Ground** tilemap is the stone floor. **Walls** has a **Tilemap Collider 2D**, a **Composite Collider 2D** and a static **Rigidbody 2D**, so the walls are solid. The tiles are in `Assets/_game/Tilemaps/Tiles/Dungeon/`: a wall cell with floor below it uses `dungeon_wall_face_bottom`, the cell above that `dungeon_wall_face_top`, and the rest `dungeon_wall_dark`. (**Fences** and **Water** are empty.)

**Characters > Player** (T01–T05, T08, T09). **TopDownMovement** (Move Speed 4), **Health** (Max Health 5), **TopDownHurt** (knockback and invincibility), **DeathEffect**, **HealthTint**, **Inventory** and **AimedShooter** (fireballs). Its **Sprite** child has **DirectionalAnimator** and **MeleeAttack**, and the **Sword Hitbox** below that has **AttackHitbox**. The **Lantern** child is a warm **Point** light (C10).

**Characters > Slime 1–6** (T06, T08). **EnemyAI** (Detection Range 4, Give Up Range 7, Chase Speed 2.5), **Health** (Max Health 3), **TopDownHurt**, **HealthTint** and **DeathEffect**. Health's **On Died** event calls **DeathEffect.Spawn**, which drops a **Coin Pickup** or a **Heart** with a **Drop Chance** of 0.6.

**Pools** (S07, T05). **Fireball Pool** (10 fireballs) and **Impact Pool** (5 impacts). The Player's **AimedShooter** takes fireballs from them instead of using Instantiate.

**Dungeon** (T10, T11, T13). The interactive pieces:

- **Locked Door** (T10): **LockedDoor** with **Key** set to the **Gold Key** asset. Bump into it carrying the key and it opens, using the key up.
- **Pressure Plate** and the first **Gate** (T10): the plate's **On Pressed** calls **Gate.Open** and **On Released** calls **Gate.Close**. The plate doesn't know about the gate; the events connect them.
- **Crate** (T10): a **Rigidbody 2D** with **Mass** 3, **Linear Damping** 10 (so it stops when you stop pushing), **Gravity Scale** 0 and rotation frozen.
- **Lever** and the second **Gate** (T10): the lever's **On Switched** calls **Gate.Toggle**.
- **Old Adventurer** (T11): **Talkable** with five **Lines**. They show in the **Dialogue Canvas**'s **DialogueBox**, which types 40 letters a second and turns off the Player's movement and attacks while you talk.
- **Chest 1**, **Chest 2**, **Treasure Chest** (T13): **Chest**, each with a **Chest Id**, **Contents** (Potion, Gold Key, Blue Gem) and **Amount** (the treasure chest gives 5).

**Rooms** (T12). Each room has a **Box Collider 2D** trigger the size of the room and **CameraRoom**. Its child is a **Cinemachine Camera** with a **Confiner 2D** that keeps the view inside the room's box. While the player's feet are in a room, that room's camera gets the higher priority. On the **Main Camera**, the **Cinemachine Brain**'s **Default Blend** is **Ease In Out**, **0.8** seconds, which is the smooth slide between rooms.

**HUD Canvas** (S09, T09): the **Health Bar** and the **Inventory** row. **Global Light 2D** (C10): **Intensity** 0.35, which makes the dungeon dim. **Treasure Save** (T13): remembers opened chests.

The one change to old code is in **TreasureSave**. It gained two fields:

```
    // A different file for each game or level, so their chest ids don't mix (T14 uses
    // its own).
    [SerializeField] private string fileName = "treasure.json";
    // Untick to start fresh every time you press Play (T14 does: its doors aren't saved,
    // so a restart would lock the door again after the key had been used).
    [SerializeField] private bool keepBetweenPlays = true;
```

Here **File Name** is `dungeon.json` and **Keep Between Plays** is off. Doors and gates aren't saved, so if you used the key and played again, the door would be locked, the key spent and its chest already open: you'd be stuck. A trap like that is called a **softlock**.

**Test it:** tick **Keep Between Plays**, open a chest, stop and play again: the chest stays open. Right-click the **Treasure Save** component's title and choose **Delete Save** to reset, then untick it again.

## 3. Build your own

Pick tasks from this list. Each one changes one system, and none needs new code except the last two. Save the scene often.

1. **More hints.** Add lines to the **Old Adventurer**'s **Lines** list. Then make a second talkable character or sign: duplicate the Old Adventurer (Ctrl+D), move it into another room and change its **Speaker Name** and **Lines**.
2. **A tougher slime.** Pick one slime and give it **Max Health** 5, **Chase Speed** 3.5 and **Detection Range** 6. Make it bigger (**Scale** 1.5) so the player can tell it apart. Is the room still fair?
3. **A different room change.** On the **Main Camera**'s **Cinemachine Brain**, set **Default Blend** to **Cut**, then to 2 seconds. Which feels best, and why?
4. **More light or less.** Change the **Global Light 2D Intensity** and the **Lantern**'s outer radius and colour. A darker dungeon is scarier but harder to read.
5. **A second key and door.** In `Assets/_game/ScriptableObjects/Items/`, duplicate **Gold Key** and make a **Silver Key**: change its **Item Name**, and give it a new **Icon** if you import one (only the gold key picture is in this project). Duplicate the **Locked Door**, move it into a doorway and set its **Key** to the Silver Key. Put the key in a chest: duplicate a chest, give it a new **Chest Id** and set its **Contents**. Add the Silver Key to **Treasure Save**'s **All Items** list.
6. **A new room.** Paint a new room with the dungeon tiles (make a Tile Palette from `Tilemaps/Tiles/Dungeon/`, as in C03), with floor on **Ground** and walls on **Walls**. Duplicate a **Room** object, move it over the new room and resize its **Box Collider 2D** to fit (at least 20 × 11.25 units, the camera's view). Drag the new room's child camera into its **Room Camera** slot. Then fill the room: slimes, pickups, a chest.
7. **Harder: save the doors.** Make a locked door remember that it was opened, the way **Chest** asks **TreasureSave**, so **Keep Between Plays** can be ticked without the softlock.
8. **Harder: a new puzzle.** A gate that needs *two* pressure plates pressed at once. You'll need a small new script that both plates call.

## Check your work

- ☐ You can play the dungeon from the entrance to the treasure chest.
- ☐ You found each sample's piece in the Hierarchy and can say which sample it came from.
- ☐ At least two of the **Build your own** tasks work, and the dungeon can still be finished.
- ☐ Compare with a fresh copy of `CRE132_T14_TopDownSlice_Finished` if you break something.

## Key points

- **A game is many small systems** that don't know about each other. The plate doesn't know about the gate, the chest doesn't know about the UI: Unity Events, C# events and public methods connect them.

- **Prefabs and data assets make content cheap.** A new key, chest or slime is a duplicate with different Inspector values, not new code.
- **Saving only some things can break a game.** Whatever the save remembers must agree with everything it forgets.

## If something goes wrong

- **The camera doesn't change in your new room:** the room's **Box Collider 2D** doesn't cover where the player's feet are, or its **Room Camera** slot still points at the old room's camera.
- **The player walks through your new walls:** you painted them on **Ground** instead of **Walls**.
- **A duplicated chest is already open or empty when you play:** it has the same **Chest Id** as another chest, and **Keep Between Plays** is ticked. Give each chest its own id and **Delete Save**.
- **The new key doesn't open the door** (the Console says `Locked. You need a ...`): the door's **Key** slot is a different asset from the one in the chest.

## Try it

1. Swap T06's slimes for the NavMesh version from your T07 project, so they follow you through doorways (bake the NavMesh from the **Walls** tilemap).
2. Add a boss room: one big slime with lots of health, guarding the treasure.
3. Give the dungeon a goal: show a "You win" message (S03) when the treasure chest is opened.
4. Add a second level and go to it through a doorway, using T12's **SpawnDoor** and **SpawnPoint** scripts (copy them from your T12 project) and the **Scene Loader**.